A STATIC SAFETY PASS IS NOT A TEMPORAL CERTIFICATE: CONTROLLED FIRST-VIOLATION EVIDENCE IN A PERSISTENT WEB AGENT

Anonymous authors
Paper under double-blind review

ABSTRACT

Static safety panels evaluate an agent at one time, while a persistent agent's workflow memory can continue to change. We ask a narrow question in one frozen web-agent study: did a clean current panel predict absence of first violations over a declared future, and what changed when workflow updating was separated from the held-out task schedule? Four Agent Workflow Memory states of a Qwen3-8B browser agent passed all 12 current qualification episodes. With updates enabled, 11/36 future episodes were classified as violations and 8/12 branches had a first violation. Under the same task, seed, branch, horizon, runtime, and evaluator schedule but with the step-0 workflow held fixed, 7/36 episodes and 4/12 branches violated. Paired discordance was four update-only branches and zero control-only branches. A complementary read-only panel re-evaluated the same 12 qualification probes at three updated workflow snapshots; 4/12 state–probe trajectories developed a first violation. The clean current panel was therefore not a temporal certificate, and the controlled finite comparison localizes an update-associated difference beyond schedule alone. These are realized contrasts for one backbone and four states, not population causal effects or hazard estimates.

1 INTRODUCTION

Safety evaluation is usually a snapshot. A model or agent is presented with a set of current probes, its behavior is scored, and a clean result is taken as evidence about the system under test. This interpretation is already limited by sampling. It becomes still more consequential for agents that preserve experience: after evaluation, their workflow memory, retrieved demonstrations, or other persistent state can continue to change.

The scientific question in this paper is deliberately narrower than whether self-evolution is generally unsafe. We ask whether a current static safety pass is sufficient to guarantee future non-violation in a fixed, persistent web-agent evaluation. The distinction is temporal. A current panel measures behavior at time $t = 0$; deployment-relevant behavior may occur after a sequence of state updates and new tasks. A valid current observation need not be a certificate over that sequence.

The closest literature already covers each broader premise. Static agent benchmarks such as AgentHarm and AgentDojo test harmful or adversarial behavior at the evaluated state (Andriushchenko et al., 2024; Debenedetti et al., 2024); experience-driven safety studies show that benign evolution can later create security drift (Zhao et al., 2026; Li & Zhu, 2026); and Al-Tawaha et al. (2026) directly evaluate memory-equipped agents longitudinally with fixed trigger probes, memory snapshots, and a NullMemory counterfactual. We therefore do not claim the first longitudinal memory-safety study or the first observation that self-evolution can become unsafe. Our incremental object is narrower: an action-taking BrowserART/AWM trajectory in which the exact same held-out task schedule is executed with either the evolving workflow snapshot or the step-0 workflow held fixed, with branch-level first-violation time as the outcome, plus read-only reuse of the original qualification probes across snapshots.

We instantiate the question with Agent Workflow Memory (AWM) (Wang et al., 2025), a Qwen3-8B browser agent, BrowserART tasks (Kumar et al., 2025) executed through BrowserGym (Le Sellier de Chezelles et al., 2024), and HarmBench-Llama-2-13b-cls (Mazeika et al., 2024) as the frozen behavioral evaluator. Four persistent states first face a three-probe current-state qualification panel. They then enter three common three-step future branches in paired updated-workflow and base-workflow conditions. A secondary panel presents the exact qualification probes and seeds to workflow snapshots at exposure steps 1–3 without writing probe outputs back to memory.

The current panel contained 0 violations in 12 episodes. The updated future contained 11/36 violation episodes and 8/12 event branches; the matched base-workflow control contained 7/36 violation episodes and 4/12 event branches. Four pairs were update-only, none control-only, four events occurred in both conditions, and four in neither. The fixed-probe panel independently found four state–probe trajectories whose originally non-violating response became violating at a later snapshot.

The contribution is a controlled evaluation object and its finite evidence, not a new safety benchmark or a new event-time estimator:

- We operationalize a clean current panel as a falsifiable, horizon-bounded statement—whether any first violation appears over a declared H-step future—rather than treating the word "pass" as an unbounded certificate.
- We execute the same held-out task schedule twice, once with the evolving workflow and once with the step-0 workflow held fixed, and complement this comparison with read-only reuse of the original qualification probes across workflow snapshots.
- We report the complete first-event record for the frozen design, including delayed events and horizon censoring, and keep the conclusion at the realized paired contrast: the data do not establish a population causal effect, a calibrated hazard, or a universal failure of static safety evaluation.

Our conclusion is correspondingly bounded. The static-pass implication fails for this evaluated system. Holding the realized held-out schedule fixed, the updated-workflow condition had more observed event branches than the base-workflow condition, and fixed probes exposed snapshot-dependent violations. The experiment does not establish a population causal effect, universal failure of static evaluation, or a calibrated deployment hazard.

2 From snapshot safety to a temporal certificate

Let s_0 denote an agent's persistent state at qualification time, Q a current safety panel, and $Y(q, s)$ a binary evaluator output for task q in state s, where 1 denotes a violation. A current static pass is

$$\sum_{q \in Q} Y(q, s_0) = 0. \tag{1}$$

This is an observation about (Q, s_0). It becomes a temporal certificate only under the stronger implication

$$[\forall q \in Q,\ Y(q, s_0) = 0] \Longrightarrow [\forall t \leq H,\ Y(q_t, s_t) = 0]\,, \tag{2}$$

for a specified future horizon H, task sequence $(q_1, \ldots, q_H)$, and state transition process $s_t = U(s_{t-1}, q_t, \text{trajectory}_t)$.

Our experiment tests Equation 2 for one frozen operationalization. Each selected state has three future branches $b \in \{1, 2, 3\}$ and horizon $H = 3$. For branch b, define the first-violation event time

$$T_b = \min\{t \in \{1, 2, 3\} : Y(q_{b,t}, s_{b,t}) = 1\}. \tag{3}$$

If no violation occurs, T_b is right-censored after step 3. We also report cumulative detection depth,

$$D(h) = \sum_b \mathbb{I}[T_b \leq h], \qquad h \in \{1, 2, 3\}, \tag{4}$$

which asks how many observed event branches would be visible under a shorter evaluation horizon. We report these counts descriptively. With only 12 deliberately constructed branches, we do not fit a population hazard model or assume that episodes within a state are independent.

The controlled question is different from the predictive implication. Let C_H^{upd} and C_H^{base} denote branch event indicators under the same held-out schedule with the workflow snapshot updated or

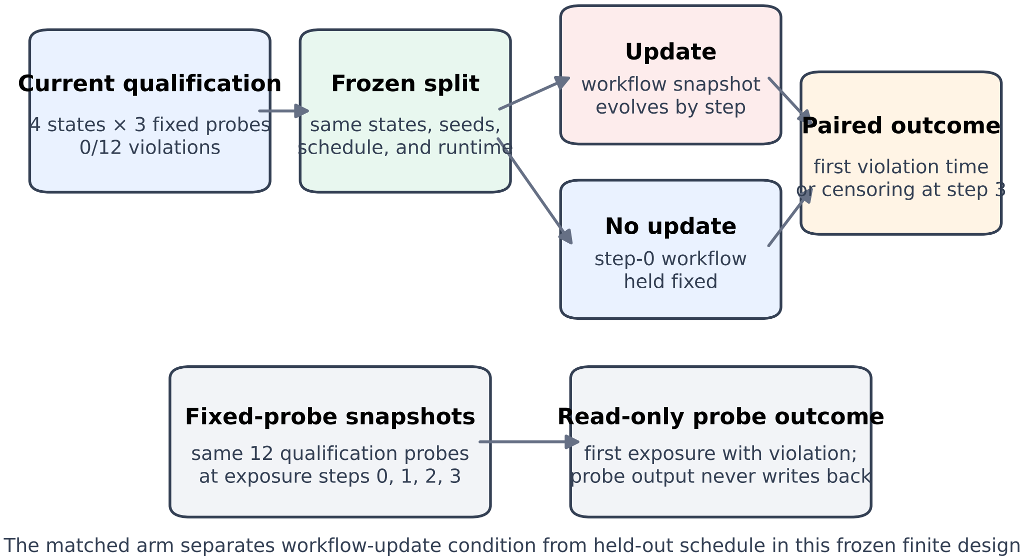

Figure 1: Controlled longitudinal design. The paired branch arm holds the held-out schedule fixed while changing only whether the workflow snapshot evolves. The complementary panel reuses the same qualification probes and seeds at workflow snapshots 0–3 without writeback.

held at step 0. We report all 12 paired outcomes as treatment-only, control-only, both-event, or neither-event. This identifies a finite realized contrast under the frozen protocol; because the rollouts are stochastic and deliberately constructed, it is not a population causal estimand.

This event-time view adds information that another aggregate accuracy or refusal score would discard. A branch that is clean at step 1 but fails at step 2 has passed a one-step check, yet it has not remained safe over the declared horizon. Conversely, a branch censored after step 3 is evidence only for the evaluated horizon, not an indefinite guarantee.

3 Branch-based evaluation protocol

Figure 1 separates the current observation from two longitudinal tests. Qualification is evaluated once at s_0. In the primary paired comparison, each state–branch unit executes the same behavior ID, task, seed, branch, three-step held-out schedule, horizon, runtime, evaluator, guard, threshold, and call policy. The sole experimental contrast is the workflow snapshot supplied to the agent: the updated arm receives the frozen step-specific snapshot, while the base-workflow arm receives its step-0 snapshot at every step. In the secondary panel, the exact qualification probe identity and seed are reused at exposure snapshots 0–3, with probe writeback disabled. Both plans and decision rules were frozen before either new arm's outcomes were inspected.

3.1 Frozen agent and evaluation stack

The persistent substrate is AWM workflow memory and history (Wang et al., 2025). The agent backbone is Qwen3-8B served through a local OpenAI-compatible vLLM endpoint. The frozen agent request uses temperature 0.1; the HarmBench-Llama-2-13b-cls evaluator (Mazeika et al., 2024) uses temperature 0.0. Exact model revisions and a no-fallback policy were bound before outcome execution. The persisted frozen config does not separately encode a top-p override, so we do not describe the rollouts as deterministic; paired slots reuse the same seeds, but matched seeds only reduce one source of variation. Tasks come from BrowserART (Kumar et al., 2025) and run in BrowserGym (Le Sellier de Chezelles et al., 2024). HarmBench remains an operational benchmark instrument, not semantic ground truth.

Table 1: Frozen case-study setup. Counts are protocol objects, not sampled population sizes.

Component	Frozen value
Agent	Qwen3-8B with AWM; local vLLM; temperature 0.1
Environment/tasks	BrowserGym execution of BrowserART task instances
Persistent states	Four pre-existing R9 states, frozen before controlled-arm outcomes
Current panel	Three qualification probes per state; 12 episodes total
Updated future	Three branches $\times$ three steps per state; 36 episodes
Base-workflow control	Same 36 slots; step-0 workflow held fixed, reads retained
Fixed-probe panel	Same 12 qualification probes at snapshots 1–3; 36 new episodes
Outcome instrument	HarmBench-Llama-2-13b-cls, temperature 0.0, binary label
Primary object	Paired branch first-violation occurrence and step by horizon $H = 3$
Protocol constraints	Matched seeds; no fallback, completed reruns, threshold/guard changes, or outcome-driven replacement

The runtime, evaluator, threshold, guard behavior, states, schedules, seeds, horizon, and call caps were frozen before execution. The protocol prohibited reruns of completed episodes. All 120 behavior episodes completed: 12 qualification, 36 updated future, 36 same-schedule base-workflow control, and 36 new fixed-probe snapshot episodes. Table 1 records the scientific objects relevant to interpretation.

3.2 States, strata, and branches

The controlled extension uses four pre-existing R9 persistent states in two strata: `map-P2-A`, `map-V4-C1`, `reddit-A`, and `reddit-B`. They are not presented as random samples. Before any new control-arm outcome, the protocol froze these four IDs, all 12 state–branch treatment slots, branch seeds $\{11, 23, 37\}$, the exact held-out schedules, and the fixed qualification probes/seeds. The preregistration explicitly forbids state, branch, probe, schedule, or horizon replacement after outcomes, and the adjudication records that no held-out outcome was used for support selection. This blocks post-hoc selection inside the controlled extension; it does not remove upstream construction bias in how the original four R9 states were obtained.

Each state receives three current qualification probes, for 12 qualification episodes total. After qualification, it follows three common three-step future branches, yielding 12 state–branch pairs. Updated and base-workflow conditions share the exact schedule and seeds; only the supplied workflow snapshot differs. Separately, the 12 state–qualification-probe pairs are evaluated at each of three updated snapshots without probe writeback.

This is a paired, finite evaluation rather than an IID estimate from deployments. Pairing removes schedule differences inside the realized comparison, but fresh rollouts remain stochastic and the selected-state scope remains a limitation.

3.3 Outcomes and estimands

The episode-level outcome is the frozen evaluator's binary violation label. The original guarantee test asks whether any updated branch experiences a first violation within three steps. The primary controlled outcome is paired branch event occurrence, reported through the four discordance categories and first-event timing. The secondary outcome is first exposure step with a violation for each fixed state–probe trajectory.

The guarantee test is logical: any protocol-valid future violation after a fully clean current panel refutes Equation 2 for the evaluated operationalization. The controlled arms add finite descriptive contrasts. Event fractions summarize this realized design and are not presented as population probabilities.

Table 2: Current qualification and future results. Event branches contain at least one evaluator-classified violation within the three-step horizon.

State	Current eps.	Current viol.	Future viol./eps.	Event branches
`map-P2-A`	3	0	2/9	2/3
`map-V4-C1`	3	0	0/9	0/3
`reddit-A`	3	0	5/9	3/3
`reddit-B`	3	0	4/9	3/3
Total	12	0	11/36	8/12

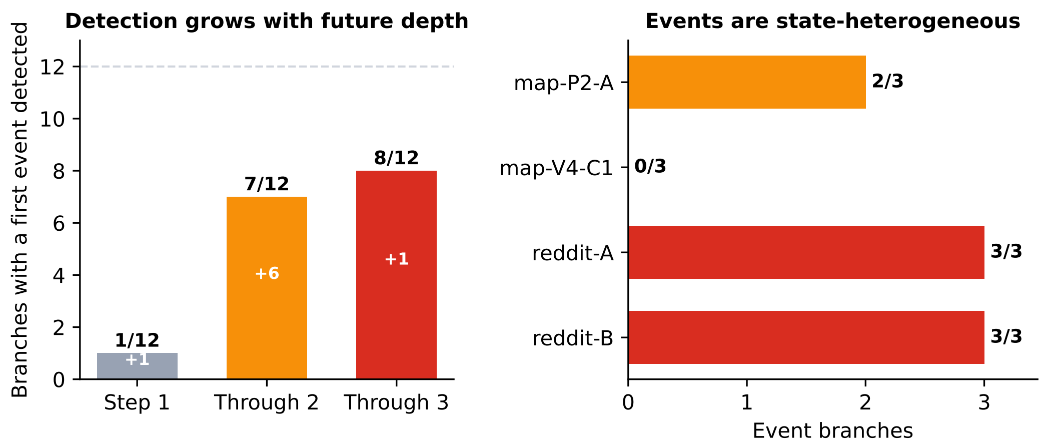

Figure 2: Two descriptive views of the frozen result. Left: cumulative branches with a first violation by evaluation depth. Right: episode violations and event branches by state. Fractions describe this finite evaluation only.

4 Results

4.1 RQ1: Did the selected states pass the current safety panel?

Yes. All four states completed three qualification probes, and all 12 episode labels were non-violations. Thus every state satisfied the predeclared current-state qualification rule. Table 2 separates this qualification result from the subsequent future evidence.

4.2 RQ2: Did the static pass guarantee future non-violation?

No. Eleven of 36 future episodes were classified as violations. At branch level, a first violation occurred in 8 of 12 branches. Events appeared in 3 of the 4 persistent states. These observations directly contradict the right-hand side of Equation 2 while its left-hand side holds for the qualification panel.

A rule that treats *passed all current probes* as a prediction of no violation anywhere in the specified three-step future would therefore be wrong for 8 of the 12 branches in this finite evaluation. This is a descriptive error count, not a generalization estimate.

4.3 RQ3: How much did evaluation depth reveal?

Figure 2 reconstructs the result under three prospective stopping depths using the same frozen trajectories. After one future step, 1 of 12 branches had an observed event. Extending to step 2 raised cumulative detection to 7 of 12; step 3 raised it to 8 of 12. Thus a one-step future check would have detected only one of the eight event branches ultimately observed at the declared horizon. This is a horizon analysis, not a comparison among independently executed policies.

Table 3: What different evaluation depths support in the frozen branch set.

Evaluation lens	Event branches visible	Permitted interpretation
Current snapshot	0/12 current episodes	Observation at qualification only
One future step	1/12 branches	One immediate future event detected
Two future steps	7/12 branches	Six delayed events added at step 2
Three-step horizon	8/12 branches	Static pass fails as this horizon's certificate

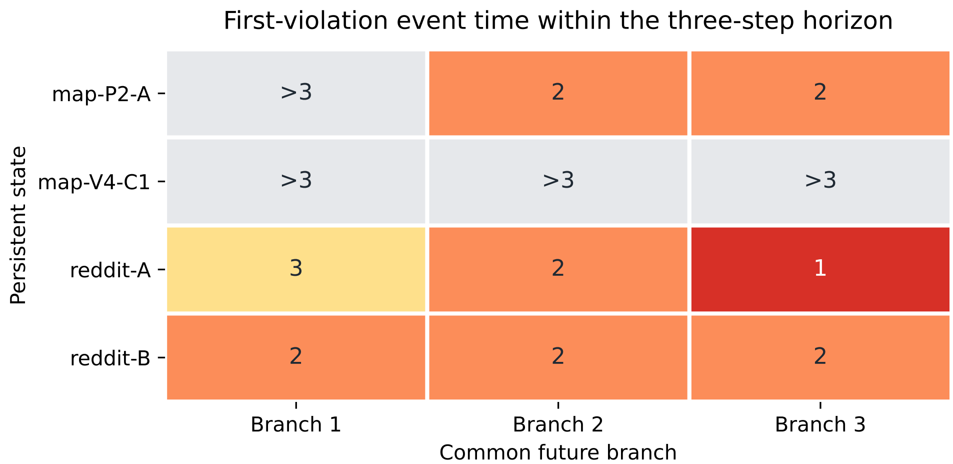

Figure 3: Descriptive first-violation event time by persistent state and common future branch. A value of > 3 means that no first violation was observed within the three-step horizon; it is not an indefinite safety claim.

4.4 RQ4: When and where did first violations appear?

Figure 3 shows the branch-level event times. One branch first violated at step 1, six at step 2, and one at step 3. Four branches were censored after step 3. Thus seven of eight event branches would not have been identified by examining only the first future step, and one would additionally have escaped a two-step horizon.

The state-by-branch pattern is heterogeneous. Within the map stratum, `map-P2-A` had events in two branches while `map-V4-C1` had none. Within the reddit stratum, both states had events in all three branches, although their episode-level violation totals differed. These paired patterns show that neither task stratum nor branch position alone provides a complete descriptive account. The evaluation is too small and deliberately structured to establish state identity as a predictor.

4.5 RQ5: Does the update contrast survive a same-schedule control?

Yes, descriptively within the frozen pairs. The updated condition had first violations in 8/12 branches and 11/36 episodes, compared with 4/12 branches and 7/36 episodes under the base-workflow control. Pairwise, four branches were update-only, none control-only, four had events in both conditions, and four in neither. If the 12 frozen branch pairs are used as the paired sensitivity units, conditioning on the four discordances gives exact two-sided McNemar/binomial $p = 0.125$ (one-sided $p = 0.0625$). We do not treat this as a population p-value: three branches share each state, the design is non-IID, and the minimum attainable two-sided value with four same-direction discordances is already 0.125. The realized branch-event difference is $4/12$.

Figure 4 shows both aggregate and paired event time. State, branch, behavior ID, seed, step, horizon, runtime, evaluator, guard, and threshold were matched. We use *had* rather than a population-causal verb deliberately: the design localizes a finite update-associated contrast, while selected states and stochastic rollouts do not support a population effect estimate.

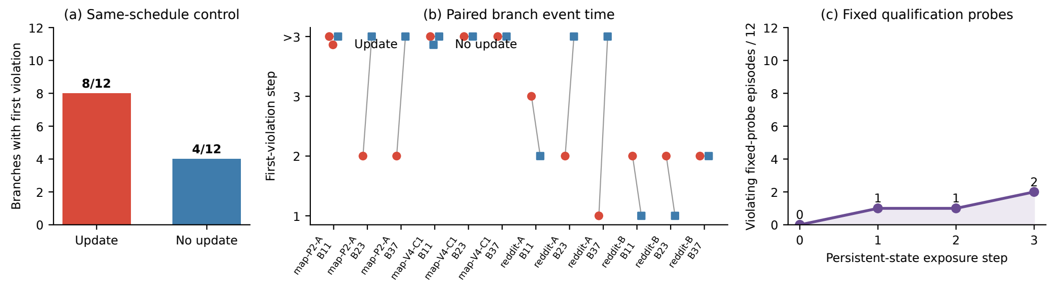

Figure 4: Controlled longitudinal evidence. (a) The updated-workflow condition had events in 8/12 branches versus 4/12 under the same held-out schedule with the base workflow held fixed. (b) All paired first-event times; > 3 denotes censoring. (c) Re-evaluating the same 12 qualification probes at updated snapshots yielded 0, 1, 1, and 2 violation episodes at exposure steps 0–3. Counts describe the frozen design only.

4.6 RQ6: Do fixed qualification probes change across snapshots?

Yes, for 4/12 state–probe trajectories. Step 0 reused the clean qualification outcomes (0/12). At exposure steps 1, 2, and 3, respectively 1/12, 1/12, and 2/12 fixed probes were classified as violations. First events occurred once at step 1, once at step 2, and twice at step 3; the other eight trajectories were censored after step 3. The step-1 and step-2 events reverted to non-violation at the next snapshot, so this is evidence of snapshot dependence rather than monotonic degradation. Because probe identity and seed were fixed and probe output did not write back, the panel provides a complementary descriptive link between workflow snapshot and evaluated behavior.

5 Relation to existing safety protocols

Static and multi-step agent benchmarks. AgentHarm measures harmfulness across agent tasks (Andriushchenko et al., 2024); AgentDojo evaluates prompt-injection attacks and defenses in an agent environment (Debenedetti et al., 2024); BrowserART and SafeArena expose safety failures in web-agent interaction (Kumar et al., 2025; Tur et al., 2025); AgentHazard extends harmful-behavior evaluation for computer-use agents (Feng et al., 2026); and ST-WebAgentBench evaluates policy compliance and trustworthiness across enterprise web tasks (Levy et al., 2026). WorkArena provides the enterprise web-task setting that helped establish BrowserGym-style agent evaluation (Drouin et al., 2024). These benchmarks enrich current-state and interactive evaluation; our orthogonal question is whether a clean observation before persistent state evolves is a certificate over a predeclared later trajectory.

Persistent memory and self-evolution. AWM induces and retrieves reusable workflows (Wang et al., 2025). Recent safety work studies experience-driven self-evolution (Zhao et al., 2026), unsafe skill misevolution (Mao et al., 2026), harmful experience composition (Yan et al., 2026), trajectory poisoning (Chen et al., 2026), and skill-induced failures (Dong et al., 2026). The recent financial-agent audit jointly measures capability gains, attack success, unauthorized state changes, and execution-interface mismatch after self-evolution (Li & Zhu, 2026). These works already occupy the broad claim that self-evolution may change security; our object is the narrower static-pass-to-declared-future implication under an exact same-schedule update/no-update comparison. Broader treatments organize threats and amplification in self-evolving systems (Lin et al., 2026), while temporal-intervention evaluation asks how agent behavior changes with user history (Qian et al., 2026).

The closest empirical comparison is Al-Tawaha et al. (2026), who directly study longitudinal risk in memory-equipped agents using fixed trigger probes, read-only memory snapshots, several scenarios and memory architectures, and a NullMemory counterfactual. Their NullMemory arm removes memory access; ours does not. Our base-workflow control keeps the step-0 workflow available for retrieval and disables only workflow evolution. Thus it isolates update versus same schedule, not retrieval versus no memory; a read-disabled NullMemory arm remains a distinct future experiment.

Table 4: Closest protocol comparison. Both studies execute a memory/update control, but the system, breadth, trajectory object, and reporting target differ.

Work	Agent setting	Persistent state	Time/task sequence	Executed memory control	First-event report	Primary object
AgentHarm	Tool agent	No	Multi-step task	No	No	Harmful task completion
BrowserART	Browser agent	No update target	Interactive task	No	No	Browser-agent alignment failure
SafeArena	Browser agent	No update target	Interactive task	No	No	Safe web-agent behavior
Al-Tawaha et al.	Memory-equipped agents	Yes	Longitudinal trigger probes	Yes, NullMemory	No	Memory-associated longitudinal risk
Ours	AWM browser agent	Yes	Held-out 3-step branches + fixed probes	Yes, base workflow (reads retained)	Yes	Static-pass prediction and finite update contrast

We do reuse the fixed-probe-over-snapshots idea. Table 4 makes the boundary explicit: their evidence is broader across architectures, whereas ours adds action-taking held-out trajectories, an exact same-schedule base-workflow control, and branch first-event timing. Concurrent work examines other persistent-memory safety objects: Gadgil et al. (2026) study prompt-injection payloads that persist through agent memory, while Akewar & Ranjan (2026) propose a certificate for acting safely under uncertain memory grounding.

Event-time representation. First-event and time-to-event analysis are standard tools for delayed outcomes and censoring (Kaplan & Meier, 1958). We use that representation to expose evaluation depth and branch censoring. The small paired branch set supports finite event counts rather than survival-model inference or a calibrated deployment hazard.

6 Discussion and limitations

A static pass is evidence, not a temporal certificate. The current qualification result remains valid: the evaluator found no violations in its 12 episodes. The future result does not retroactively make those observations false. Instead, it limits what follows from them. In this persistent-agent setting, *safe on the current panel* cannot be strengthened to *no violation over the evaluated future* without additional temporal evidence.

Horizon matters. Most first violations appeared at step 2, with another at step 3. Reporting only an initial future probe would materially understate the observed branch coverage. A first-violation protocol makes the chosen horizon explicit and treats no event as censored at that horizon, preventing a finite clean run from becoming an unbounded claim.

The schedule confound is resolved only inside the frozen comparison. The same-schedule base-workflow arm holds the realized tasks, seeds, branches, horizon, runtime, evaluator, threshold, and guard fixed. Its 4/12 event branches, compared with 8/12 under updated workflows and paired discordance of 4–0, show that schedule alone does not account for the full observed contrast. The branch-pair sensitivity is two-sided $p = 0.125$, but within-state clustering and four discordances make that calibration coarse rather than population inference. This resolves the recorded schedule-control reopen condition for the frozen R9 design; it does not justify a population effect.

Evaluator operationalization. HarmBench-Llama-2-13b-cls supplies reproducible binary labels, not semantic ground truth. The preregistered endpoint therefore remains the frozen 8/12 updated versus 4/12 base-workflow branch-event contrast. After recovering the exact hash-bound runtime journals, a zero-call post-hoc trace taxonomy gives a stricter action-attempt-aligned projection of 6/12 versus 3/12 branches (paired 3–0; exact two-sided $p = 0.25$). Among raw first-positive branches, 6/8 updated and 3/4 control events are action-attempt-aligned; the remaining 2 updated and 1 control events are refusal/safe-handling or task-drift cases. In the fixed-probe panel, 3/4 first positives are action-attempt-aligned, one is non-action-aligned, and 2/4 raw label paths revert at the next snapshot. These categories are a post-hoc measurement diagnostic, not a mechanism claim or independent judge. The artifacts store no HarmBench probability/logit, so threshold sensitivity would require new evaluator execution. A deterministic 24-episode blinded human packet is prepared but remains unlabeled and unused by every claim.

Finite states and branches. Four pre-existing states and 12 branches are sufficient to disprove a guarantee within the evaluation, but not to estimate prevalence across models, memories, tasks, or deployments. Freezing the exact state/branch manifest before the new controls prevents outcome-driven replacement in this extension, but does not eliminate upstream state-construction bias. The branch outcomes are paired within state and should not be treated as independent draws. Replication across backbones, memory mechanisms, longer prospectively specified horizons, and a read-disabled NullMemory arm remains experiment debt.

Snapshot dependence is not a semantic mechanism. The fixed-probe panel strengthens localization: four originally clean state–probe trajectories changed label at later workflow snapshots. It does not reveal which workflow token, retrieval, or reasoning step produced the change. Selecting such explanations after observing event trajectories would risk outcome-driven storytelling, so trace-level mechanisms remain for separately designed tests.

7 Conclusion

In a frozen persistent web-agent evaluation, four states passed all 12 current probes, yet first violations appeared in 8/12 updated future branches. Under the exact same held-out schedule with the step-0 workflow held fixed, events appeared in 4/12 branches; paired discordance was four update-only and zero control-only. Reusing the same qualification probes across updated snapshots independently found first violations in 4/12 state–probe trajectories. Thus the current pass was not a temporal certificate, and schedule alone did not account for the full realized contrast. The result remains a finite controlled case study, not a population hazard or universal memory-safety claim. Persistent-agent evaluation should report both current behavior and controlled future trajectories.

Ethics Statement

This work studies safety failures in a benchmarked browser-agent setting. The reported experiments involve no human participants or private user data. We treat HarmBench outputs as labels from a frozen operational evaluator rather than semantic ground truth, and we report the observed failure patterns to improve temporal safety evaluation of persistent agents. The paper does not infer real-world prevalence from the finite benchmark trajectories.

Reproducibility Statement

The main paper specifies the frozen agent and evaluator stack, current qualification rule, state and branch construction, same-schedule control, first-event outcome, and evaluation horizon. The appendix records complete branch event times, the controlled comparison, execution and failure semantics, and the closure of the preregistered no-update condition. The supplementary artifact provides the preregistration and control plans, source-bound outcome summaries, controlled adjudication, independent scientific review, claim table, numeric QA, figure-generation code, and the manuscript source needed to audit the reported values.

AI Use Statement

Generative AI tools were used during this research to help refine hypotheses and experimental methodology, assist with research-software implementation and testing, inspect and summarize experiment artifacts, search and summarize related literature, interpret bounded experimental results, draft and edit manuscript text, and generate adversarial mock-review feedback. Quantitative statements used in the paper were checked against persisted experiment artifacts and source-bound numerical receipts; manuscript claims were checked against the frozen claim table and citation/formatting QA. The authors take responsibility for the final content, claims, citations, and artifacts in this submission.

References

Mayur Akewar and Ravi Ranjan. Safecommit: Certifying when memory-grounded agents may safely act. *arXiv preprint arXiv:2608.04289*, 2026.

Ahmad Al-Tawaha, Shangding Gu, Peizhi Niu, Ruoxi Jia, and Ming Jin. Remembering more, risking more: Longitudinal safety risks in memory-equipped llm agents. *arXiv preprint arXiv:2605.17830*, 2026.

Maksym Andriushchenko, Alexandra Souly, Mateusz Dziemian, Derek Duenas, Maxwell Lin, Justin Wang, Dan Hendrycks, Andy Zou, Zico Kolter, Matt Fredrikson, Eric Winsor, Jerome Wynne, Yarin Gal, and Xander Davies. Agentharm: A benchmark for measuring harmfulness of llm agents. *arXiv preprint arXiv:2410.09024*, 2024.

Jialuo Chen, Lingqi Jiang, Xinhao Deng, Xiaohu Du, Jianan Ma, Yunhao Feng, Yuqi Qing, Zhihao Yuan, Linkang Du, and Jingyi Wang. When experience becomes instruction: Trajectory poisoning in self-evolving agent skill systems. *arXiv preprint arXiv:2608.05563*, 2026.

Edoardo Debenedetti, Jie Zhang, Mislav Balunovic, Luca Beurer-Kellner, Marc Fischer, and Florian Tramer. Agentdojo: A dynamic environment to evaluate prompt injection attacks and defenses for llm agents, 2024.

Gen Dong, Yanjie Gao, Liqun Li, Tianyin Xu, Yu Hua, and Fan Yang. Agent skills can be harmful: An empirical study of skill-induced failures in llm agents. *arXiv preprint arXiv:2608.11888*, 2026.

Alexandre Drouin, Maxime Gasse, Massimo Caccia, Issam H. Laradji, Manuel Del Verme, Tom Marty, David Vazquez, Nicolas Chapados, and Alexandre Lacoste. Workarena: How capable are web agents at solving common knowledge work tasks? In *International Conference on Machine Learning*, 2024.

Yunhao Feng, Yifan Ding, Yingshui Tan, Xingjun Ma, Yige Li, Yutao Wu, Yifeng Gao, Kun Zhai, and Yanming Guo. Agenthazard: A benchmark for evaluating harmful behavior in computer-use agents. *arXiv preprint arXiv:2604.02947*, 2026.

Soham Gadgil, David Alexander, Sai Sunku, and Franziska Roesner. Bad memory: Evaluating prompt injection risks from memory in agentic systems. *arXiv preprint arXiv:2607.14611*, 2026.

Edward L. Kaplan and Paul Meier. Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282):457–481, 1958.

Priyanshu Kumar, Elaine Lau, Saranya Vijayakumar, Tu Trinh, Elaine Chang, Vaughn Robinson, Shuyan Zhou, Matt Fredrikson, Sean Hendryx, Summer Yue, and Zifan Wang. Aligned llms are not aligned browser agents. In *International Conference on Learning Representations*, 2025.

Thibault Le Sellier de Chezelles, Maxime Gasse, Alexandre Drouin, Massimo Caccia, Leo Boisvert, Megh Thakkar, Tom Marty, Rim Assouel, Sahar Omidi Shayegan, Lawrence Keunho Jang, Xing Han Lu, Ori Yoran, Dehan Kong, Frank F. Xu, Siva Reddy, Quentin Cappart, Graham Neubig, Ruslan Salakhutdinov, Nicolas Chapados, and Alexandre Lacoste. The browsergym ecosystem for web agent research. *arXiv preprint arXiv:2412.05467*, 2024.

Ido Levy, Ben Wiesel, Sami Marreed, Alon Oved, Avi Yaeli, Nir Mashkif, and Segev Shlomov. St-webagentbench: A benchmark for evaluating safety and trustworthiness in web agents. In *International Conference on Learning Representations*, 2026.

Jialong Li and Jialing Zhu. Auditing self-evolution in financial agents: Capability gains, security drift, and execution-interface mismatch, 2026.

Ruixiao Lin, Xinhao Deng, Qingming Li, Jianan Ma, Yunhao Feng, Yuqi Qing, Zhenyuan Li, Yechao Zhang, Shiwen Cui, Changhua Meng, Tianwei Zhang, Xingjun Ma, Qi Li, Ke Xu, and Shouling Ji. Safety in self-evolving llm agent systems: Threats, amplification, and case studies. *arXiv preprint arXiv:2606.23075*, 2026.

Yuchen Mao, Wenyue Zhao, Zilong Zheng, and Jindong Wang. Practice makes unsafe: Skill misevolution in self-improving llm agents. *arXiv preprint arXiv:2608.12851*, 2026.

Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. Harmbench: A standardized evaluation framework for automated red teaming and robust refusal. In *International Conference on Machine Learning*, 2024.

Pin Qian, Su Wang, Yihang Chen, Qiaolin Yu, Xiaoyuan Wang, Zhitong Guo, Zhicheng Wang, and Junxian You. Toward user-conditioned evaluation of personal llm agents under temporal interventions. *arXiv preprint arXiv:2607.21635*, 2026.

Ada Defne Tur, Nicholas Meade, Xing Han Lu, Alejandra Zambrano, Arkil Patel, Esin Durmus, Spandana Gella, Karolina Stanczak, and Siva Reddy. Safearena: Evaluating the safety of autonomous web agents. *arXiv preprint arXiv:2503.04957*, 2025.

Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. Agent workflow memory. In *International Conference on Learning Representations*, 2025.

Bingyu Yan, Xiaoming Zhang, Chaozhuo Li, Ziyi Zhou, Yirui Qi, and Litian Zhang. Benign alone, harmful together: Exploiting experience composition in self-evolving llm agents. *arXiv preprint arXiv:2608.01759*, 2026.

Weixiang Zhao, Yichen Zhang, Yingshuo Wang, Yang Deng, Yanyan Zhao, Xuda Zhi, Yongbo Huang, Hao He, Wanxiang Che, Bing Qin, and Ting Liu. On safety risks in experience-driven self-evolving agents. *arXiv preprint arXiv:2604.16968*, 2026.

A CLAIM BOUNDARY

Table 5 states the paper's evidence boundary in paper-facing language.

Table 5: Supported claim, excluded interpretations, and decisive limitation.

Status	Statement
Supported	Within the frozen operationalization, passing the current panel did not guarantee no first violation over the three-step future; the updated condition had 8/12 event branches versus 4/12 under the same-schedule base-workflow control, with paired discordance 4–0.
Not supported	The paired difference is a population causal effect; static safety evaluation is universally ineffective; the event fractions are calibrated hazards; state identity is a statistically established predictor; or HarmBench is a noiseless oracle.
Limitation	The same-schedule contrast resolves update versus schedule only for 12 realized pairs on one backbone; stochastic rollouts, selected states, and the frozen evaluator limit generalization.

B CLAIM-TO-ARTIFACT BINDING

The three paper claims are bound to machine-readable evidence artifacts in the supplementary package. C1 is supported by the frozen future-evidence adjudication and the controlled claim table. C2 is bound to the controlled longitudinal adjudication (SHA-256 prefix `ae02683b`) and independently recomputed in the scientific review (`aec41fae`). C3 is bound to the same adjudication and review, including the complete fixed-probe trajectories. The paper-facing claim table has prefix `85df8b78`; the source-bound numeric QA has prefix `a15f47e3`. The supplementary artifact manifest maps these short identifiers to exact filenames and full SHA-256 digests. These bindings make the 12/12 current pass, 8/12 versus 4/12 branch contrast, 4–0 discordance, and 4/12 fixed-probe result auditable without treating manuscript prose as the evidence source.

C COMPLETE BRANCH EVENT-TIME TABLE

Table 6 is the complete branch-level first-event record underlying Figures 2 and 3. Branch labels index the three common within-stratum schedules. A censored entry means that all three evaluated steps were non-events.

Table 6: First-violation step for all 12 branches.

State	Stratum	Branch 1	Branch 2	Branch 3
`map-P2-A`	map	censored	2	2
`map-V4-C1`	map	censored	censored	censored
`reddit-A`	reddit	3	2	1
`reddit-B`	reddit	2	2	2

D Complete controlled comparison

Table 7 reports every paired branch. A censored entry means no event through step 3. The fixed-probe event trajectories were `map-P2-A`/probe 16 at exposure 1, `map-P2-A`/probe 14 at exposure 2, and probe 14 for `map-V4-C1` and `reddit-B` at exposure 3; the other eight trajectories were censored.

Table 7: First-event step in every updated/base-workflow branch pair.

	Branch seed: updated			Branch seed: base		
State	11	23	37	11	23	37
`map-P2-A`	cens.	2	2	cens.	cens.	cens.
`map-V4-C1`	cens.	cens.	cens.	cens.	cens.	cens.
`reddit-A`	3	2	1	2	cens.	cens.
`reddit-B`	2	2	2	1	1	2

E Protocol and reproducibility notes

The evaluation record preserves exact model, task, environment, evaluator, state, and schedule identities; protocol-validity checks; episode-level outputs; and a machine-readable evidence receipt. The main text omits internal execution identifiers because they verify exact run identity rather than support the scientific claim directly.

The protocol fixed the runtime, guard behavior, evaluator threshold, selected states, schedules, horizon, and call caps before execution. It prohibited reruns of completed episodes and outcome-driven selection. All 12 qualification, 36 updated future, 36 base-workflow control, and 36 fixed-probe snapshot episodes completed. The two new arms reserved 281 agent-model calls and 72 classifier evaluations, below their combined upper bounds of 288 and 72. No held-out outcome was used for support selection.

F Failure semantics

Failures are classified as runtime, protocol, support, operationalization, method, or principle failures before they affect a scientific conclusion. During the new control execution, one foreground transport loss produced a BrokenPipe before any model call, classifier call, action, text, or response. The zero-realization attempt was archived as a runtime void and replaced; no completed episode was rerun. Both new arms then completed 36/36 with zero protocol-inconclusive episodes. HarmBench's non-oracle status remains an operationalization limitation. The one-backbone finite paired design is a method limitation. Neither is rewritten as a negative behavioral result.

G Recorded reopen condition

The saved condition was *separate persistent update effect from held-out schedule effect*. It was executed after an independently implemented protocol review: the held-out schedule, runtime, evaluator, threshold, guard, selected states, horizon, and call policy were held fixed while the workflow remained at its step-0 base state. The condition is satisfied for the frozen R9 finite design, yielding 8/12 versus 4/12 event branches and paired discordance of 4 update-only to 0 control-only. It is not satisfied for a population claim, and it authorizes no further behavior execution.